

Erez Arie Zimmerman

Observational Astrophysicist | Supernovae | Time-Domain Astronomy

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Research Interests

Infant supernovae; supernova progenitors; circumstellar material; flash spectroscopy; stellar mass loss; thermonuclear explosions; red supergiants; UV and X-ray astronomy; time-domain surveys; transient discovery and follow-up; astronomical data systems.

Research Profile and Leadership

I am an observational astronomer studying infant supernovae, circumstellar interaction, and the late-stage evolution of massive stars. My work combines rapid multi-wavelength follow-up from the ground, UV, and X-rays with transient survey infrastructure and target-of-opportunity coordination.

Supernovae from the ground and from space

My research focuses on discovering and observing supernovae from the first hours after explosion to late phases. I am interested in constraining the properties of supernova **progenitor stars**, mapping their **mass-loss** history, and understanding supernova **explosion mechanisms**. In particular, I have studied nearby supernovae, including the decadal supernova **2023ixf** (Zimmerman et al. 2024, published in *Nature*), for which I obtained *HST* UV spectra, and the extremely faint Iax supernova **2024vjm** (Zimmerman et al., in review for *Nature Astronomy*), for which I analyzed a pre-explosion *Euclid* space telescope image containing the explosion site. I have led several successful telescope proposals with instruments such as the **Hubble Space Telescope**, **Gemini**, **VLT**, and others.

Collaborations

I am an active participant in major transient collaborations including **LAST**, **BlackGEM**, **ePESSTO+**, **SoXS**, and formerly **ZTF**, detecting, classifying and following-up interesting Supernovae. I lead the BlackGEM consortium bi-weekly transient meetings, regularly participate in ePessto+ classifications as part of the TNS team, and developed **CAST**, a TOM-based transient marshal for the LAST collaboration.

Education

Weizmann Institute of Science

Ph.D. in Astrophysics; advisor: Prof. Avishay Gal-Yam

Rehovot, Israel

2021 – expected Oct. 2026

Weizmann Institute of Science

M.Sc. in Astrophysics; advisor: Prof. Avishay Gal-Yam

Rehovot, Israel

2018 – 2021

Technion – Israel Institute of Technology

B.Sc. in Physics

Haifa, Israel

2015 – 2018

First-Author Publications

The Complex Circumstellar Environment of Supernova 2023ixf

Erez A. Zimmerman, Ido Irani, Ping Chen, Avishay Gal-Yam et al.

Nature, 627, 759–762, 2024

A Faint Progenitor System for the Faint Supernova 2024vjm

Erez A. Zimmerman, Avishay Gal-Yam et al.

Nature Astronomy, submitted and in review

Accepted Telescope Time and Observing Proposals

Infant Supernovae: Observing Massive Stars as They Explode with the Large Array Survey Telescope
Gemini 2025B
4.5 hours

Observing the Local Environment of the Nearby Faintest Iax SN 2024vjm with MUSE Narrow Field Mode
ESO Period 114 DDT
2.9 hours, VLT/MUSE

Observing the Circumstellar Interaction of the 7 Mpc SN 2024ggi Chandra DDT 25508981
30 ks

High-Resolution Spectroscopy of One of the Nearest Supernovae in a Decade: SN 2024ggi ESO Period 113
DDT
2.7 hours, VLT/UVES

Explosions in Real Time: Rapid UV Supernova Flash Spectroscopy HST Cycle 30
22 orbits

Infant Supernovae: Observing Massive Stars as They Explode OPTICON 2024B
Liverpool Telescope 5 hr + TNG 0.3 nights

Infant Supernovae: Observing Massive Stars as They Explode from the Ground and Space Gemini 2023A
0.9 nights

Explosions in Real Time: Rapid UV Supernova Flash Spectroscopy OPTICON 2023A
TNG 1.0 night

Observational Experience

- **Hubble Space Telescope:** 22 orbits.
- **VLT/MUSE:** 1 remote observing run.
- **VLT/UVES:** 4 remote observing runs.
- **NTT, La Silla Observatory, Chile:** 9.5 nights.
- **Lowell Discovery Telescope, Lowell Observatory, Arizona, USA:** 1 night as well as ToO observations.
- **Neil Gehrels Swift Observatory:** 40+ approved ToO observations
- **Aristarchos Telescope, Chelmos Observatory, Greece:** 6 nights.
- **Target-of-opportunity activations:** extensive experience triggering rapid ToO photometry and spectroscopy with multiple facilities and instruments.

Honors and Awards

Weizmann School of Science Prof. Israel Dostrovsky Prize of Excellence	2026
Weizmann School of Science Haim Holtzman Memorial Prize	2025
Best Poster, 3rd place, 67th Meeting of the Israel Physical Society	2022

Teaching

Lecturer, Open University of Israel	2022 – 2026
<i>Secrets of the Universe: A Window to Astronomy</i>	

Instructor for a general astronomy course covering the history of astronomy, the Solar System, astronomical instrumentation, stellar physics, supernovae and supernova remnants, galaxy formation, and cosmology. Average student rating: 4.3/5.

Invited Talks

SN 2023ixf Early Emission and Circumstellar-Material Mapping	Jun. 2024
Review talk, SN 2023ixf: The Decadal Supernova in M101 Workshop, European Southern Observatory, Garching, Germany.	

Ultraviolet Observations Resolve the Explosion of Supernova 2023ixf in Messier 101 within Its Complex Circumstellar Environment	Jan. 2024
American Astronomical Society Meeting 243, New Orleans, LA, USA.	

Dying Stars Shed Light on Late-Stage Mass Loss	Nov. 2025
Invited seminar, Technion, Haifa, Israel.	

A Framework to Map Circumstellar Material Around Core-Collapse Supernovae	May 2025
Invited seminar, Hebrew University of Jerusalem, Jerusalem, Israel.	

The Complex Circumstellar Environment of Supernova 2023ixf	Feb. 2024
Tel Aviv University Student Seminar, Tel Aviv University, Tel Aviv, Israel.	

Selected Contributed Talks

A Faint Progenitor System for the Faint Supernova 2024vjm	Jun. 2026
American Astronomical Society Meeting 248, Pasadena, CA, USA.	

Dying Stars Shed Light on Late-Stage Mass Loss	Nov. 2025
Stellar Populations and Their Explosions: Bridging the Gap Workshop, Max Planck Institute for Astrophysics, Ringberg Castle, Germany.	

SN 2024vjm: A Faint Supernova and a Fainter Companion	Sep. 2025
BlackGEM Consortium Fall Meeting 2025, Hamburg Observatory, Hamburg, Germany.	

A Framework to Map the Circumstellar Material Around Red Supergiant Supernovae	Jul. 2025
Israel Physical Society Meeting, Technion, Haifa, Israel.	

Ultraviolet Observations of Infant Supernovae	Mar. 2025
Transients from Space, STScI, Baltimore, MD, USA.	

SN 2024ggi and Infant Supernovae: Early Light Curve and Circumstellar Material Sep. 2024
BlackGEM Consortium Fall Meeting 2024, University of Manchester, Manchester, UK.

Early and UV Observations of SN 2023ixf, a Core-Collapse Supernova in the Nearby M101 Galaxy Aug. 2023
SNeX – SuperNova EXplosions Conference, Technion, Haifa, Israel.

High-Velocity Circumstellar Wind Around the Type II Supernova SN 2020pni Jun. 2022
Zwicky Transient Facility Spring Meeting 2022, CNRS/IN2P3, Paris, France.

Selected Posters

A Faint Progenitor System for the Faint Supernova 2024vjm Aug. 2024
IAU Symposium 406 – Future Landscape of Astrophysical Transients, Turku, Finland.

The Complex Circumstellar Environment of Supernova 2023ixf Aug. 2024
International Astronomical Union General Assembly, Cape Town, South Africa, remote.

High-Velocity Circumstellar Wind Around the Type II Supernova SN 2020pni Feb. 2022
Israel Physical Society 67th Meeting, Ben-Gurion University, Be'er Sheva, Israel.

Outreach, Service, and Media

- Recorded lecture for Israeli Space Week, *Telescopes in Space Reveal the Mysteries of the Universe*, for ages 13–14, Hebrew Jan. 2025
- Guest lecture for the Israeli Astronomical Society, *We Are All Stardust: Supernovae and the Origin of the Elements*, Hebrew Sep. 2024
- Guest, *Voice of the Universe* podcast, episode 1, Hebrew Apr. 2024
- Guest, BBC *Science in Action*, “Brightest Supernova in a Decade” May 2023
- Press coverage of Zimmerman et al. 2024 in [Weizmann](#), [Stockholm University](#), [Keck Observatory](#), [Ynet](#), [Mako](#), and Kan Radio.

Other Professional Experience and Affiliations

Weizmann–Helmholtz Multi-Messenger School Fellow 2021 – present
Summer Student, CERN Jun. 2018 – Sep. 2018

Selected Co-Author Publications

The Type II In SN 2025cbj coincidence with the high-energy neutrino IceCube-250421A
Simone Garrappa, Erez A. Zimmerman, et al.
[A&A](#), 708 (2026)

XSNAP: An X-ray Supernova Analysis Pipeline with Application to the Type II Supernova 2024ggi
Ferdinand, Wynn V. Jacobson-Galán, Mansi M. Kasliwal, Erez A. Zimmerman
[ApJ](#), 1001, 26 (2026)

Extremely Stripped Supernova Reveals a Silicon and Sulfur Formation Site

Steve Schulze et al.

Nature, 2024, 644, pages634–639 (2025)

A Search for Minute-timescale Flares from the Transient AT 2024wpp

Eran O. Ofek et al.

ApJ, 993, 76 (2025)

The Progenitor Star of SN 2023ixf: A Massive Red Supergiant with Enhanced, Episodic Pre-Supernova Mass Loss

Yu-Jing Qin, Keming Zhang, Joshua Bloom, Jesper Sollerman, Erez A. Zimmerman, et al.

MNRAS, 534, Issue 1, 271–280 (2024)

The BlackGEM Telescope Array. I. Overview

Paul J. Groot et al.

PASP, 136, 115003 (2024)

The Early Ultraviolet Light Curves of Type II Supernovae and the Radii of Their Progenitor Stars

Ido Irani et al.

ApJ, 970, 96 (2024)

The Prevalence and Influence of Circumstellar Material Around Hydrogen-rich Supernova Progenitors

Rachel J. Bruch, Avishay Gal-Yam, Ofer Yaron, Ping Chen, Nora L. Strotjohann, Ido Irani, Erez Zimmerman, et al.

ApJ, 952, 119 (2023)

A WC/WO Star Exploding Within an Expanding Carbon–Oxygen–Neon Nebula

Avishay Gal-Yam et al.

Nature, 601, 201–204 (2022)